



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Explainable AI-Driven Multimodal Analytics for Sustainable Smart City Decision Intelligence	
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Abstract

Rapid urbanization has resulted in several challenges, including increasing air pollution, unpredictable weather conditions, and rising energy consumption. These factors have a significant impact on environmental sustainability and urban planning. Managing these interconnected issues efficiently has become a major challenge for developing sustainable smart cities.

Existing smart city applications mainly focus on monitoring individual factors such as air quality, weather, or energy consumption separately. However, they often lack an integrated approach and provide limited insights into the relationship between these domains. This creates the need for a unified and explainable data analytics system that supports informed decision-making.

The proposed project, "Explainable Data Analytics for Sustainable Smart City Applications," integrates Air Quality, Historical Weather, and Energy Consumption datasets to analyze urban sustainability. The system performs data preprocessing, exploratory data analysis, and predictive modeling using Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. SHAP (SHapley Additive Explanations) is used to improve the transparency and interpretability of the model predictions.

In the future, the proposed system can be extended by integrating real-time IoT sensor data, supporting multiple cities, and incorporating advanced deep learning techniques to enhance prediction accuracy. It can also be expanded into a comprehensive Smart City Decision Support System for sustainable urban planning and efficient resource management.

Signature of the Guide

HOD